



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

10-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	C	C	D	B	B	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	D	B	A	A	A	B	A	B

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	D	D	A	C	B	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	D	B	C	A	D	C	B	A

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	D	C	B	A	A	D	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	B	B	C	D	B	A	A

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

$$qE = Mg$$

$$neE = \rho \left(\frac{4}{3} \pi r^3 \right) \times g$$

$$n \times 1.6 \times 10^{-19} \times 3.55 \times 10^5$$

$$= 3 \times 10^3 \times \frac{4}{3} \times \pi \times (2 \times 10^{-3})^3 \times 9.81$$

$$n = 173 \times 10^{(3-9-5+19)}$$

$$n = 1.73 \times 10^{10}$$

2. Ans (A)

Order of atmosphere stratification from bottom Troposphere, stratosphere, Mesosphere, Thermosphere

(a) → (iv)

(b) → (iii)

(c) → (ii)

(d) → (i)

3. Ans (B)

$$E \propto \frac{1}{r} \quad r \propto \frac{1}{m}$$

$$E \propto m$$

$$\text{Ionization potential} = 13.6 \times \frac{(\text{Mass}_\mu) \text{ eV}}{(\text{Mass}_e)}$$

$$= 13.6 \times 207 \text{ eV} = 2815.2 \text{ eV}$$

4. **Ans (C)**

$$E = BC = 6$$

$$(\text{Dir. of wave}) \parallel (\vec{E} \times \vec{B})$$

$$\hat{i} = \hat{j} \times \hat{k}$$

$$\vec{E} = 6\hat{j} \text{ V/m}$$

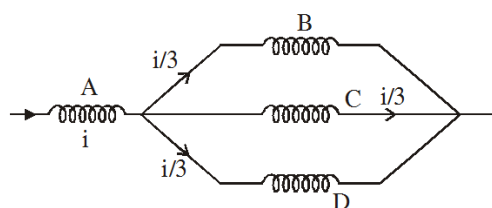
5. **Ans (C)**

Using conservation of angular momentum

$$(Mr^2)\omega = (Mr^2 + 2mr^2)\omega'$$

$$\omega' = \frac{M\omega}{M + 2m}$$

6. **Ans (D)**



$$\phi \propto i$$

$$\Rightarrow B \propto i$$

$$\text{so, field at centre of C} = \frac{3}{3} = 1 \text{ T}$$

7. **Ans (B)**

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + 2 \frac{\Delta T}{T} = \frac{0.1}{10} + 2 \left(\frac{\frac{1}{200}}{0.5} \right)$$

$$\frac{\Delta g}{g} = \frac{1}{100} + \frac{1}{50}$$

$$\frac{\Delta g}{g} \times 100 = 3\%$$

8. **Ans (B)**

$$P = C$$

$$FV = C$$

$$M \frac{dV}{dt} V = C$$

$$\frac{V^2}{2} \propto t$$

$$V \propto t^{1/2}$$

$$\frac{dx}{dt} \propto t^{1/2}$$

$$x \propto t^{3/2}$$

9. **Ans (A)**

Energy associated with each degree of freedom

$$\text{per molecule} = \frac{1}{2} k_B T$$

10. **Ans (C)**

$$\lambda_{eq} = \lambda_1 + \lambda_2$$

$$\frac{1}{T_{1/2}} = \frac{1}{T_{1/2}^{(1)}} + \frac{1}{T_{1/2}^{(2)}}$$

$$T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$$

11. **Ans (A)**

Adiabatic process is from C to D

$$W_D = \frac{P_2 V_2 - P_1 V_1}{1 - \gamma}$$

$$= \frac{P_D V_D - P_C V_C}{1 - \gamma}$$

$$= \frac{200(3) - (100)(4)}{1 - 1.4}$$

$$= -500 \text{ J}$$

12. **Ans (B)**

$$\beta = \frac{\lambda D}{d} = \frac{5890 \times 10^{-10} \times 0.5}{0.5 \times 10^{-3}}$$

$$= 589 \times 10^{-6} \text{ m}$$

Distance between first and third bright fringe is

$$2\beta = 2 \times 589 \times 10^{-6} \text{ m}$$

$$= 1178 \times 10^{-6} \text{ m}$$

13. **Ans (D)**

$$\lambda = \frac{h}{p}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{p_e}{p_p} = \frac{m_e v_e}{m_p v_p}$$

$$2 = \frac{m_e}{m_p} \left(\frac{v_e}{4v_e} \right)$$

$$\therefore m_p = \frac{m_e}{8}$$

14. **Ans (B)**

Option (B) represent correct graph for particle moving with constant acceleration, as for constant acceleration velocity time graph is straight line with positive slope and x-t graph should be an opening upward parabola.

15. **Ans (A)**

$$R = \frac{\rho \ell}{A} = \frac{V}{I}$$

$$\rho = \frac{AV}{I\ell} = \frac{\pi d^2 V}{4I\ell} \left(A = \frac{\pi d^2}{4} \right)$$

$$\therefore \frac{\Delta \rho}{\rho} = \frac{2\Delta d}{d} + \frac{\Delta V}{V} + \frac{\Delta I}{I} + \frac{\Delta \ell}{\ell}$$

$$\frac{\Delta \rho}{\rho} = 2 \left(\frac{0.01}{5.00} \right) + \frac{0.1}{5.0} + \frac{0.01}{2.00} + \frac{0.1}{10.0}$$

$$\frac{\Delta \rho}{\rho} = 0.004 + 0.02 + 0.005 + 0.01$$

$$\frac{\Delta \rho}{\rho} = 0.039$$

$$\% \text{ error} = \frac{\Delta \rho}{\rho} \times 100 = 0.039 \times 100 = 3.90\%$$

16. **Ans (A)**

$$\text{Bandwidth} = R/L$$

$$\text{Bandwidth} \propto R$$

So bandwidth will increase

17. **Ans (A)**

$$i = i_0 \cos(\omega t)$$

$$i = i_0 \text{ at } t = 0$$

$$i = \frac{i_0}{\sqrt{2}} \text{ at } \omega t = \frac{\pi}{4}$$

$$t = \frac{\pi}{4\omega} = \frac{\pi}{4(2\pi f)} = \frac{1}{8f}$$

$$t = \frac{1}{400} = 2.5 \text{ ms}$$

18. **Ans (B)**

If distant objects are blurry then problem is Myopia. If objects are distorted then problem is Astigmatism

19. **Ans (A)**

Every part ($d\ell$) of the wire is pulled by force $i(d\ell)B$ acting perpendicular to current & magnetic field giving it a shape of circle.

20. **Ans (B)**

$$T^2 \propto R^3$$

$$\left(\frac{T'}{T} \right)^2 = \left(\frac{9R}{R} \right)^3$$

$$T'^2 = T^2 \times 9^3$$

$$T' = T \times 3^3$$

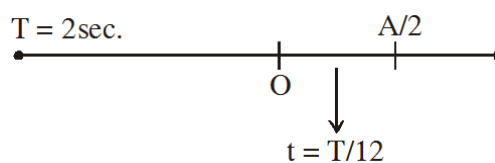
$$T' = 27 T$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (6)**

$$T = 2 \text{ sec.}$$



$$t = \frac{2}{12} = \frac{1}{6}$$

$$\therefore \text{Correct answer} = 6.00$$

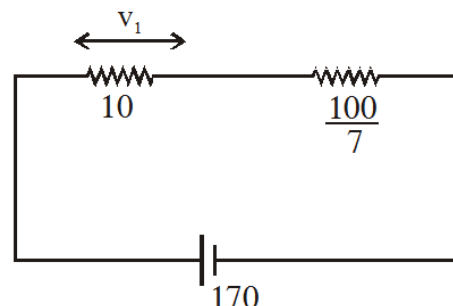
2. **Ans (2)**

$$i_0 = \frac{V}{R} = \frac{30/3}{5 \times 10^6} = 2 \times 10^{-6}$$

$$\therefore \text{Ans.} = 2.00$$

3. **Ans (70)**

$$R_{eq1} = \frac{50 \times 20}{70} = \frac{100}{7}$$



$$R_{eq} = \frac{170}{7}$$

$$V_1 = \left[\frac{170}{\frac{170}{7}} \right] \times 10 = 70 \text{ V}$$

$$\text{Ans.} = 70.00$$

4. **Ans (32)**

$$\text{For A } \frac{E}{\pi r^2} = y \frac{2\text{mm}}{a} \dots\dots(1)$$

$$\text{For B } \frac{E}{\pi \cdot 16r^2} = y \frac{4\text{mm}}{b} \dots\dots(2)$$

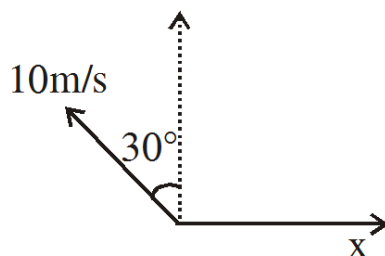
$$\therefore (1)/(2)$$

$$16 = \frac{2b}{4a}$$

$$\frac{a}{b} = \frac{1}{32}$$

$$\therefore \text{Answer} = 32$$

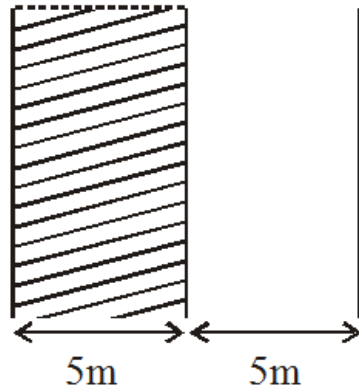
5. **Ans (5)**



$$10 \sin 30^\circ = x$$

$$x = 5 \text{ m/s}$$

6. **Ans (161)**



$$A = 100 \text{ m}^2$$

$$\text{Using } C = \frac{k\epsilon_0 A}{d}$$

$$C_1 = \frac{10\epsilon_0 (100)}{5}$$

$$= 200 \epsilon_0$$

$$C_2 = \frac{\epsilon_0 (100)}{5} = 20\epsilon_0$$

$$C_1 \text{ \& } C_2 \text{ are in series so } C_{\text{eqv.}} = \frac{C_1 C_2}{C_1 + C_2}$$

$$= \frac{4000\epsilon_0}{220}$$

$$= 160.9 \times 10^{-12} \simeq 161 \text{ pF}$$

7. **Ans (20)**

Let velocity of 2nd fragment is $\vec{v}$ then by conservation of linear momentum

$$10 (10\sqrt{3}) \hat{i} = (10) (10\hat{j}) + 10\vec{v}$$

$$\Rightarrow \vec{v} = 10\sqrt{3}\hat{i} - 10\hat{j}$$

$$|\vec{v}| = \sqrt{300 + 100} = \sqrt{400} = 20 \text{ m/s}$$

8. **Ans (10)**

Using work energy theorem,

$$W_g = \Delta \text{K.E.}$$

$$(10) (g) (5) = \frac{1}{2} (10)v^2 - 0$$

$$v = 10 \text{ m/s}$$

9. **Ans (100)**

$$10^6 = \beta^2 \times \frac{R_0}{R_i}$$

$$10^6 = \beta^2 \times \frac{10^4}{10^2}$$

$$\beta^2 = 10^4 \Rightarrow \beta = 100$$

10. **Ans (10)**

$$v^2 = u^2 + 2as$$

$$0 = (10)^2 + 2(-a)\left(\frac{1}{2}\right)$$

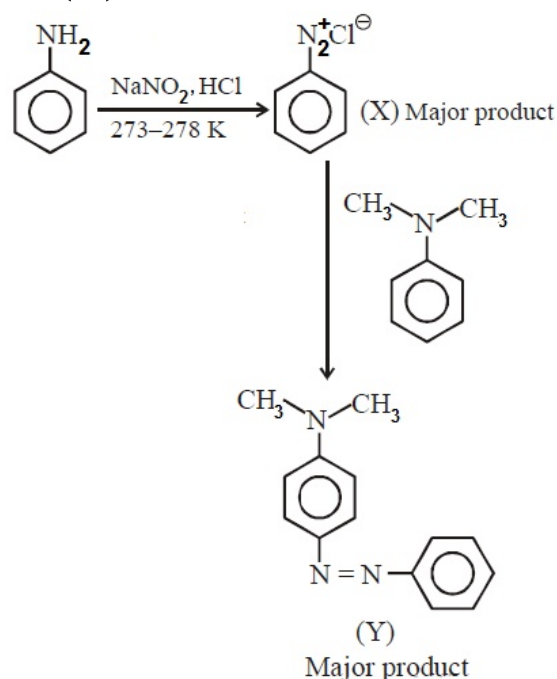
$$a = 100 \text{ m/s}^2$$

$$F = ma = (0.1) (100) = 10 \text{ N}$$

PART-2 : CHEMISTRY

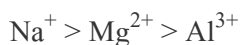
SECTION-I

1. **Ans (B)**



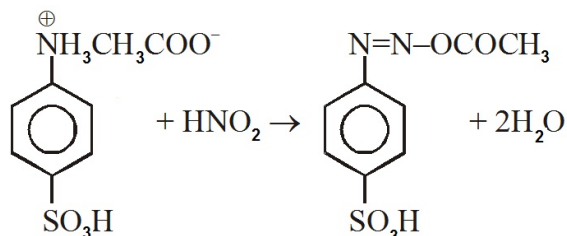
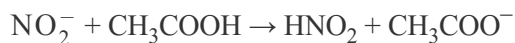
2. **Ans (B)**

The ionic radii order is

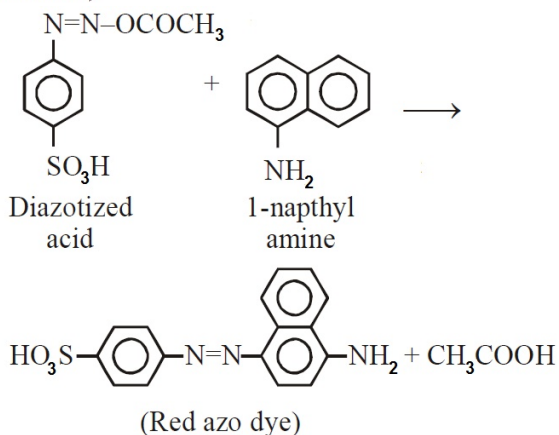


4. **Ans (D)**

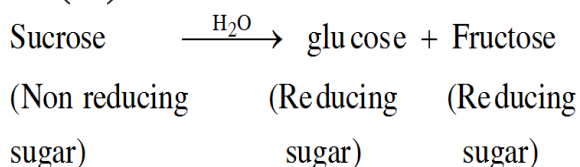
For detection of NO_2^- , the following test is used.



(Sulphanilic acid solution)



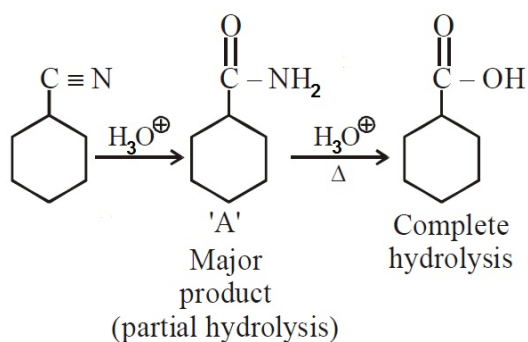
5. **Ans (D)**



6. **Ans (A)**

- (a) Antacid : Cimetidine
(b) Artificial Sweetener : Alitame
(c) Antifertility : Novestrol
(d) Tranquilizers : Valium

7. **Ans (C)**



8. **Ans (B)**

Chlorophyll is a coordination compound of magnesium.

Vitamin B-12, cyanocobalamin is a coordination compound of cobalt.

Cisplatin is used as an anti-cancer drug and is a coordination compound of platinum.

Grubbs catalyst is a compound of Ruthenium.

9. **Ans (B)**

- (a) Alcoholic potassium hydroxide $\rightarrow$ used for β -elimination
(b) Pd / BaSO₄ $\rightarrow$ Lindlar's catalyst
(c) BHC (Benzene hexachloride) $\rightarrow$ Obtained by addition reactions
(d) Polyacetylene $\rightarrow$ Electrodes in batteries

10. **Ans (A)**

Methane leads to both global warming & photochemical smog.

Methane is generated in large amounts from paddy fields.

CO₂ can be absorbed by photosynthesis, or by formation of acid rain etc., while no such activities are there for methane.

Hence methane is stronger global warming gas than CH₄.

Methane is not a part of reducing smog.

11. **Ans (C)**

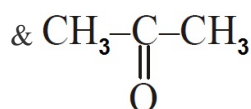
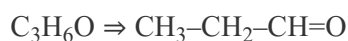
Ca(OCl)₂ is Bleach.

CaSO₄ · $\frac{1}{2}$ H₂O is plaster of paris.

CaCO₃ is used as an antacid.

CaO is major component of cement.

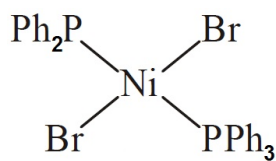
12. **Ans (D)**



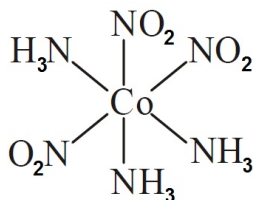
They are functional group isomerism.

13. **Ans (D)**

trans-[Ni Br₂(PPh₃)₂] is



meridional - [Co(NH₃)₃(NO₂)₃] is



14. **Ans (B)**

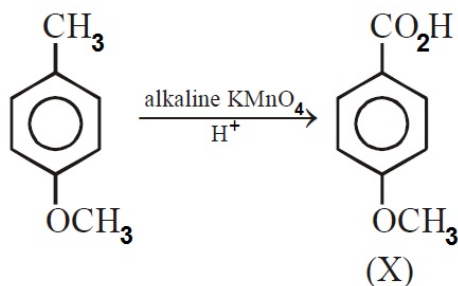
$l = 0 \Rightarrow$'s' orbital

$n - l - 1 = 2$

$n - 1 = 2$

$n = 3$

15. **Ans (C)**



16. **Ans (A)**

In manufacture of H₂SO₄ (contact process), V₂O₅ is used as a catalyst.

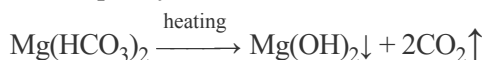
Ni catalyst enables the hydrogenation of fats.

CuCl₂ is used as catalyst in Deacon's process.

ZSM-5 used as catalyst in cracking of hydrocarbons.

17. **Ans (D)**

For temporary hardness,



Assertion is false.

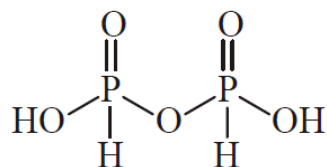
MgCO₃ has high solubility product than Mg(OH)₂.

According to data of NCERT table 7.9 (Equilibrium chapter), the solubility product of magnesium carbonate is 3.5×10^{-8} and solubility product of Mg(OH)₂ is 1.8×10^{-11} .

Hence Reason is incorrect.

The question should be Bonus.

18. **Ans (C)**



(Two ionisable H)

19. **Ans (B)**

$$M_{12 \times \frac{2}{3}} A_6$$

$$M_8 A_6$$

$$M_4 A_3$$

20. **Ans (A)**

To reduce the melting point of reaction mixture, cryolite is added.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (15)**

AX is a covalent diatomic molecule.

The molecule is NO.

Total no. of electrons is 15.

2. **Ans (10)**

$$\text{pH} = \text{pKa} + \log \frac{[\text{CB}]}{[\text{WA}]}$$

$$5.74 = 4.74 + \log \frac{[\text{CB}]}{1}$$

$$\Rightarrow [\text{CB}] = 10 \text{ M}$$

3. **Ans (3)**

$$r = k[\text{NO}]^m [\text{Cl}_2]^n$$

$$= k(0.1)^m (0.1)^n \dots (1)$$

$$= k(0.1)^m (0.2)^n \dots (2)$$

$$= k(0.2)^m (0.2)^n \dots (3)$$

$$n = 1$$

$$m = 2$$

$$m + n = 3$$

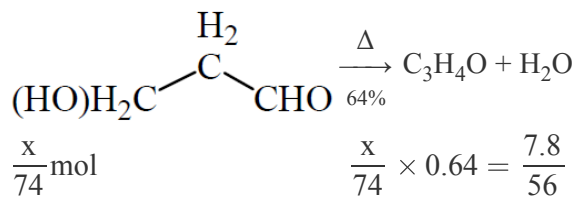
4. **Ans (128)**

$$\Delta_r H = [\epsilon_{\text{C-C}} + 2\epsilon_{\text{C-H}}] - [\epsilon_{\text{C=C}} + \epsilon_{\text{H-H}}]$$

$$= [347 + 2 \times 414] - [611 + 436]$$

$$= 128$$

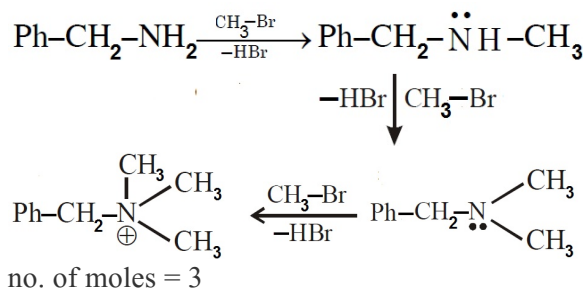
5. Ans (16)



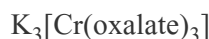
$$x = 16.10$$

$$\approx 16.00$$

6. Ans (3)



7. Ans (3)



Chromium is in +3 oxidation state.

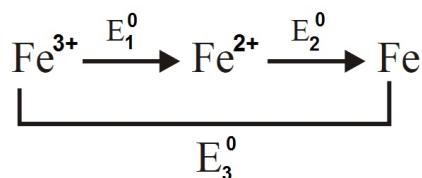
Number of unpaired electrons in Cr^{+3} will be 3.

8. Ans (50)

$$\Delta T_f = (1 + \alpha) K_f m$$

$$\alpha = 0.05 = 50 \times 10^{-3}$$

9. Ans (46)



$$E_1^0 + 2E_2^0 = 3E_3^0$$

$$E_1^0 = 3E_3^0 - 2E_2^0$$

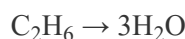
$$= 3(-0.036) - 2(-0.44)$$

$$= +0.772 \text{ V}$$

$$E_{\text{cell}}^0 = E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 + E_{\text{I}^-/\text{I}_2}^0 = 0.233$$

$$\Delta_r G^0 = -2 \times 96.5 \times 0.233 = -45 \text{ kJ}$$

10. Ans (18)



$$0.1 \quad 0.3 = 0.3 \times 6 \times 10^{23} = 18 \times 10^{22}$$

$$\text{mol} \quad \text{mol}$$

$$\text{No. of molecules} = 0.3 \times 6.023 \times 10^{23}$$

$$= 18.069 \times 10^{22}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$y^2 = 4ax + 4a^2$$

differentiate with respect to x

$$\Rightarrow 2y \frac{dy}{dx} = 4a$$

$$\Rightarrow a = \left(\frac{y}{2} \frac{dy}{dx} \right)$$

so, required differential equation is

$$y^2 = \left(4 \times \frac{y}{2} \frac{dy}{dx} \right) x + 4 \left(\frac{y}{2} \frac{dy}{dx} \right)^2$$

$$\Rightarrow y^2 \left(\frac{dy}{dx} \right)^2 + 2xy \left(\frac{dy}{dx} \right) - y^2 = 0$$

$$\Rightarrow y \left(\frac{dy}{dx} \right)^2 + 2x \left(\frac{dy}{dx} \right) - y = 0$$

2. Ans (B)

$$3x + 4y = 9$$

$$y = mx + 1$$

$$\Rightarrow 3x + 4mx + 4 = 9$$

$$\Rightarrow (3 + 4m)x = 5$$

$\Rightarrow x$ will be an integer when

$$3 + 4m = 5, -5, 1, -1$$

$$\Rightarrow m = \frac{1}{2}, -2, -\frac{1}{2}, -1$$

so, number of integral values of m is 2

3. Ans (B)

$$(1 + x + 2x^2)^{20} = a_0 + a_1x + \dots + a_{40}x^{40} \text{ put } x = 1, -1$$

$$\Rightarrow a_0 + a_1 + a_2 + \dots + a_{40} = 2^{20}$$

$$a_0 - a_1 + a_2 - \dots + a_{40} = 2^{20}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{39} = \frac{4^{20} - 2^{20}}{2}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{37} = 2^{39} - 2^{19} - a_{39}$$

$$\text{here } a_{39} = \frac{20!(2)^{19} \times 1}{19!} = 20 \times 2^{19}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{37} = 2^{19}(2^{20} - 1 - 20)$$

$$= 2^{19}(2^{20} - 21)$$

4. **Ans (D)**

$$\begin{vmatrix} 1 + \sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1 + \cos^2 x & \cos^2 x \\ 4 \sin 2x & 4 \sin 2x & 1 + 4 \sin 2x \end{vmatrix} = 0$$

use $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow (2 + 4\sin 2x) \begin{vmatrix} 1 & 1 & 1 \\ \cos^2 x & 1 + \cos^2 x & \cos^2 x \\ 4 \sin 2x & 4 \sin 2x & 1 + 4 \sin 2x \end{vmatrix} = 0$$

$$\Rightarrow \sin 2x = -\frac{1}{2}$$

$$\Rightarrow 2x = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$x = \frac{\pi}{2} + \frac{\pi}{12}, \pi - \frac{\pi}{12}$$

5. **Ans (C)**

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$A(5,5), R_1 = 3$$

$$x^2 + y^2 - 22x - 10y + 137 = 0$$

$$B(11,5), R_2 = 3$$

$$AB = 6 = R_1 + R_2$$

Touch each other externally

$\Rightarrow$ circles have only one meeting point.

6. **Ans (B)**

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0$$

$$\Rightarrow -(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \sum \alpha\beta) = 0$$

$$\Rightarrow -(-a)(a^2 - 2b - b) = 0$$

$$\Rightarrow a(a^2 - 3b) = 0$$

$$\Rightarrow a^2 = 3b \Rightarrow \frac{a^2}{b} = 3$$

7. **Ans (A)**

$$\int \frac{(2x-1) \cos \sqrt{(2x-1)^2 + 5}}{\sqrt{(2x-1)^2 + 5}} dx$$

$$(2x-1)^2 + 5 = t^2$$

$$2(2x-1) 2dx = 2t dt$$

$$2\sqrt{t^2 - 5} dx = t dt$$

$$\text{So } \int \frac{\sqrt{t^2 - 5} \cos t}{2\sqrt{t^2 - 5}} dt = \frac{1}{2} \sin t + c$$

$$= \frac{1}{2} \sin \sqrt{(2x-1)^2 + 5} + c$$

8. **Ans (A)**

$$y = mx + c$$

$$3 = m + c$$

$$\sqrt{2} = \left| \frac{m - 3\sqrt{2}}{1 + 3\sqrt{2}m} \right|$$

$$= 6m + \sqrt{2} = m - 3\sqrt{2}$$

$$= 5m = -4\sqrt{2} \Rightarrow m = \frac{-4\sqrt{2}}{5}$$

$$= 6m - \sqrt{2} = m - 3\sqrt{2}$$

$$= 7m - 2\sqrt{2} \Rightarrow m = \frac{2\sqrt{2}}{7}$$

$$\text{According to options take } m = \frac{-4\sqrt{2}}{5}$$

$$\text{So } y = \frac{-4\sqrt{2}x}{5} + \frac{3 + 4\sqrt{2}}{5}$$

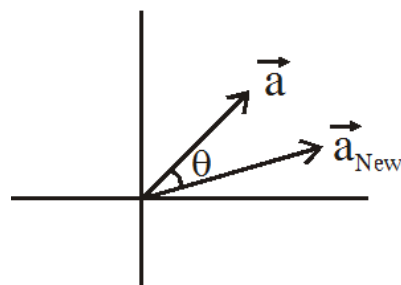
$$4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$$

9. **Ans (D)**

$$\lim_{x \rightarrow 0} \frac{\left(x + \frac{x^3}{3!} \dots\right) - \left(x - \frac{x^3}{3!} \dots\right)}{3x^3} = \frac{1}{6}$$

$$\text{So } 6L + 1 = 2$$

10. **Ans (D)**



$$\vec{a}_{\text{Old}} = 3p\hat{i} + \hat{j}$$

$$\vec{a}_{\text{New}} = (p+1)\hat{i} + \sqrt{10}\hat{j}$$

$$\Rightarrow |\vec{a}_{\text{Old}}| = |\vec{a}_{\text{New}}|$$

$$\Rightarrow ap^2 + 1 = p^2 + 2p + 1 + 10$$

$$8p^2 - 2p - 10 = 0$$

$$4p^2 - p - 5 = 0$$

$$(4p-5)(p+1) = 0 \rightarrow p = -1$$

11. **Ans (B)**

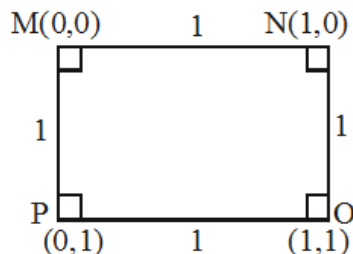
$$az\bar{z} + \alpha\bar{z} + \bar{\alpha}z + d = 0 \rightarrow \text{Circle}$$

$$\text{centre} = \frac{-\alpha}{a} \quad 2 = \sqrt{\frac{\alpha\bar{\alpha}}{a^2} - \frac{d}{a}} = \sqrt{\frac{\alpha\bar{\alpha} - ad}{a^2}}$$

$$\text{So } |\alpha|^2 - ad > 0 \text{ \& } a \in \mathbb{R} - \{0\}$$

12. **Ans (B)**

$$\begin{aligned} M : x^2 + y^2 &= 1 \quad (0,0) \\ N : x^2 + y^2 - 2x &= 0 \quad (1,0) \\ O : x^2 + y^2 - 2x - 2y + 1 &= 0 \quad (1,1) \\ P : x^2 + y^2 - 2y &= 0 \quad (0,1) \end{aligned}$$



13. **Ans (B)**

$$\begin{aligned} S &= (100)(100) + (99)(101) + (98)(102) + \dots \\ &\dots (2)(198) + (1)(199) \end{aligned}$$

$$\begin{aligned} S &= \sum_{x=0}^{99} (100-x)(100+x) = \sum 100^2 - x^2 \\ &= 100^3 - \frac{99 \times 100 \times 199}{6} \end{aligned}$$

$$\begin{aligned} \alpha &= 3 \quad \beta = 1650 \\ \text{slope} &= \frac{1650}{3} = 550 \end{aligned}$$

14. **Ans (B)**

$$f(x) = \frac{\cos e^{-1}x}{\sqrt{\{x\}}}$$

$$\text{Domain} \in (-\infty, -1] \cup [1, \infty)$$

$$\{x\} \neq 0 \text{ so } x \neq \text{integers}$$

15. **Ans (B)**

$$\begin{aligned} T_n &= \frac{1}{(2n+1)^2 - 1} \cdot \frac{1}{(2n+2)2n} = \frac{1}{4(n)(n+1)} \\ &= \frac{(n+1) - n}{4n(n+1)} = \frac{1}{4} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ S &= \frac{1}{4} \left(1 - \frac{1}{101} \right) = \frac{1}{4} \left(\frac{100}{101} \right) = \frac{25}{101} \end{aligned}$$

16. **Ans (C)**

$$f(x) + g(x) = \sqrt{x} + \sqrt{1-x}, \text{ domain } [0, 1]$$

$$f(x) - g(x) = \sqrt{x} - \sqrt{1-x}, \text{ domain } [0, 1]$$

$$g(x) - f(x) = \sqrt{1-x} - \sqrt{x}, \text{ domain } [0, 1]$$

$$\frac{f(x)}{g(x)} = \frac{\sqrt{x}}{\sqrt{1-x}}, \text{ domain } [0, 1]$$

$$\frac{g(x)}{f(x)} = \frac{\sqrt{1-x}}{\sqrt{x}}, \text{ domain } (0, 1]$$

$$\text{So, common domain is } (0, 1)$$

17. **Ans (D)**

$$f(x) = \begin{cases} \frac{1}{|x|}, & |x| \geq 1 \\ ax^2 + b, & |x| < 1 \end{cases}$$

at $x = 1$ function must be continuous

$$\text{So, } 1 = a + b \quad \dots(1)$$

differentiability at $x = 1$

$$\left(-\frac{1}{x^2} \right)_{x=1} = (2ax)_{x=1}$$

$$\Rightarrow -1 = 2a \Rightarrow a = -\frac{1}{2}$$

$$(1) \Rightarrow b = 1 + \frac{1}{2} = \frac{3}{2}$$

18. **Ans (B)**

$$A + 2B = \begin{pmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{pmatrix} \quad \dots(1)$$

$$2A - B = \begin{pmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\Rightarrow 4A - 2B = \begin{pmatrix} 4 & -2 & 10 \\ 4 & -2 & 12 \\ 0 & 2 & 4 \end{pmatrix} \quad \dots(2)$$

$$(1) + (2) \Rightarrow 5A = \begin{pmatrix} 5 & 0 & 10 \\ 10 & -5 & 15 \\ -5 & 5 & 5 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ -1 & 1 & 1 \end{pmatrix} \text{ and } 2A = \begin{pmatrix} 2 & 0 & 4 \\ 4 & -2 & 6 \\ -2 & 2 & 2 \end{pmatrix}$$

$$\therefore B = \begin{pmatrix} 2 & 0 & 4 \\ 4 & -2 & 6 \\ -2 & 2 & 2 \end{pmatrix} - \begin{pmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 1 & -1 \\ 2 & -1 & 0 \\ -2 & 1 & 0 \end{pmatrix}$$

$$\text{tr}(A) = 1 - 1 + 1 = 1$$

$$\text{tr}(B) = -1$$

$$\text{tr}(A) = 1 \text{ and } \text{tr}(B) = -1$$

$$\therefore \text{tr}(A) - \text{tr}(B) = 2$$

19. Ans (A)

Digits are 1, 2, 2, 3

$$\text{total distinct numbers} = \frac{4!}{2!} = 12.$$

total numbers when 1 at unit place is 3.

2 at unit place is 6

3 at unit place is 3.

$$\begin{aligned}\text{So, sum} &= (3 + 12 + 9) (10^3 + 10^2 + 10 + 1) \\ &= (1111) \times 24 \\ &= 26664\end{aligned}$$

20. Ans (A)

$$\text{Let } x = 3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$$

$$\text{So, } x = 3 + \frac{1}{4 + \frac{1}{x}} = 3 + \frac{1}{\frac{4x+1}{x}}$$

$$\Rightarrow (x-3) = \frac{x}{(4x+1)}$$

$$\Rightarrow (4x+1)(x-3) = x$$

$$\Rightarrow 4x^2 - 12x + x - 3 = x$$

$$\Rightarrow 4x^2 - 12x - 3 = 0$$

$$x = \frac{12 \pm \sqrt{(12)^2 + 12 \times 4}}{2 \times 4} = \frac{12 \pm \sqrt{12(16)}}{8}$$

$$= \frac{12 \pm 4 \times 2\sqrt{3}}{8} = \frac{3 \pm 2\sqrt{3}}{2}$$

$$x = \frac{3}{2} \pm \sqrt{3} = 1.5 \pm \sqrt{3}$$

But only positive value is accepted

$$\text{So, } x = 1.5 + \sqrt{3}$$

PART-3 : MATHEMATICS

SECTION-II

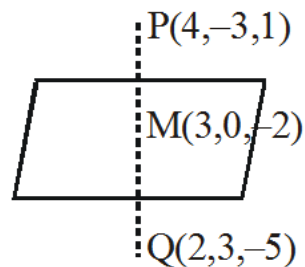
1. Ans (300)

$$3_ = 10 \times 10 = 100$$

$$_3_ = 10 \times 10 = 100$$

$$_ _ 3 = 10 \times 10 = \frac{100}{300}$$

2. Ans (28)



$$\text{Plane is } 1(x-3) - 3(y-0) + 3(z+2) = 0$$

$$x - 3y + 3z + 3 = 0$$

$$(a^2 + b^2 + c^2 + d^2)_{\min} = 28$$

3. Ans (512)

$$I = 2 \int_0^4 f(x^2) dx \quad \{\text{Even function}\}$$

$$= 2 \int_0^4 (4x^3 - g(4-x)) dx$$

$$= 2 \left(\left. \frac{4x^4}{4} \right|_0^4 - \int_0^4 g(4-x) dx \right)$$

$$= 2(256 - 0) = 512$$

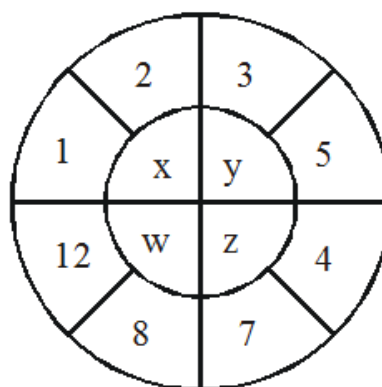
4. Ans (4)

$$x = (2-1)^{1!} = 1$$

$$w = (12-8)^{4!} = 4^{24}$$

$$z = (7-4)^{3!} = 3^6$$

$$\text{hence } y = (5-3)^{2!} = 2^2$$



5. Ans (6)

If $0, z_1, z_2$ are vertices of equilateral triangles

$$\Rightarrow a^2 + z_1^2 + z_2^2 = 0 (z_1 + z_2) + z_1 z_2$$

$$\Rightarrow (z_1 + z_2)^2 = 3z_1 z_2$$

$$\Rightarrow a^2 = 3 \times 12$$

$$\Rightarrow |a| = 6$$

6. Ans (4)

Let plane is $x - 2y + 2z + \lambda = 0$

distance from $(1, 2, 3) = 1$

$$\Rightarrow \frac{|\lambda + 3|}{5} = 1 \Rightarrow \lambda = 0, -6$$

$$\Rightarrow a = 1, b = -2, c = 2, d = -6 \text{ or } 0$$

$$b - d = 4 \text{ or } -2, c - a = 1$$

$$\Rightarrow K = 4 \text{ or } -2$$

7. Ans (35)

$$\frac{\sum x_i}{25} = 40 \& \frac{\sum x_i - 60 + N}{25} = 39$$

Let age of newly appointed teacher is N

$$\Rightarrow 1000 - 60 + N = 975$$

$$\Rightarrow N = 35 \text{ years}$$

8. Ans (4)

$$f(x) = \int \frac{(5x^8 + 7x^6) dx}{x^{14}(x^{-5} + x^{-7} + 2)^2}$$

$$\text{Let } x^{-5} + x^{-7} + 2 = t$$

$$(-5x^{-6} - 7x^{-8})dx = dt$$

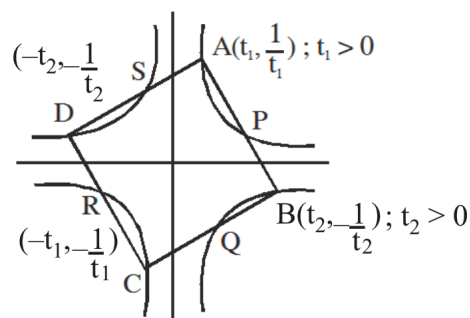
$$\Rightarrow f(x) = \int -\frac{dt}{t^2} = \frac{1}{t} + c$$

$$f(x) = \frac{x^7}{x^2 + 1 + 2x^7}$$

$$f(1) = \frac{1}{4}$$

9. Ans (80)

$$xy = 1, -1$$



$$\frac{t_1 + t_2}{2} \cdot \frac{\frac{1}{t_1} - \frac{1}{t_2}}{2} = 1$$

$$\Rightarrow t_1^2 - t_2^2 = 4t_1t_2$$

$$\frac{1}{t_1^2} \times \left(-\frac{1}{t_2^2}\right) = -1 \Rightarrow t_1t_2 = 1$$

$$\Rightarrow (t_1t_2)^2 = 1 \Rightarrow t_1t_2 = 1$$

$$t_1^2 - t_2^2 = 4$$

$$\Rightarrow t_1^2 + t_2^2 = \sqrt{4^2 + 4} = 2\sqrt{5}$$

$$\Rightarrow t_1^2 = 2 + \sqrt{5} \Rightarrow \frac{1}{t_1^2} = \sqrt{5} - 2$$

$$AB^2 = (t_1 - t_2)^2 + \left(\frac{1}{t_1} + \frac{1}{t_2}\right)^2$$

$$= 2\left(t_1^2 + \frac{1}{t_1^2}\right) = 4\sqrt{5} \Rightarrow \text{Area}^2 = 80$$

10. Ans (1)

$$\text{If } \cot x > 0 \Rightarrow \frac{1}{\sin x} = 0 \text{ (Not possible)}$$

$$\text{If } \cot x < 0 \Rightarrow 2\cot x + \frac{1}{\sin x} = 0$$

$$\Rightarrow 2\cos x = -1$$

$$\Rightarrow x = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} \text{ (reject)}$$